

Requirement Analysis Document

UniTrack BD – University Bus Tracking System

Project Type: University Transportation Management and Tracking System

Project Name: UniTrack BD

1. Introduction

UniTrack BD is a smart university bus tracking and management system designed to improve transportation efficiency, safety, and convenience for students, drivers, and university administrators. The system provides real-time bus tracking, live seat availability, QR-based payments, vehicle monitoring, and administrative control functions.

2. Project Objectives

- Provide real-time bus location tracking.
- Improve student convenience through live seat monitoring.
- Enable cashless transportation payments using QR codes.
- Increase vehicle safety through speed monitoring.
- Monitor fuel consumption and operational efficiency.
- Improve route compliance and administrative oversight.

3. Stakeholders

- Students
- Bus Drivers
- University Transport Authority
- System Administrators
- University Management

4. Functional Requirements

4.1 Real-Time Bus Tracking

- Track bus location using smartphone GPS or IoT GPS device.
- Display live location on a map.
- Update location periodically.

4.2 QR-Based Payment System

- Students can pay transportation fees using QR codes.
- Payment confirmation will be generated automatically.
- Transaction records will be stored securely.

4.3 Live Seat Availability

- Display available and occupied seats in real time.

- Students can check bus occupancy before boarding.

4.4 Vehicle Speed Monitoring

- Monitor and record vehicle speed continuously.
- Notify administrators when speed limits are exceeded.

4.5 City Map Integration

- Implement digital map support (initially Dhaka City).
- Show routes, current location, and bus stops.

4.6 Fuel Monitoring System

- Display current fuel level.
- Estimate fuel consumption during trips.
- Generate fuel usage reports.

4.7 Bus Stop Management

- Mark all stoppages between trip origin and destination.
- Display upcoming stops.

4.8 Unauthorized Route Detection

- Detect when a bus leaves the approved route.
- Generate alerts to administrators.
- Optional engine control integration for advanced implementations.

4.9 Road Monitoring System

- Monitor route conditions.
- Record road incidents or route deviations where applicable.

5. Non-Functional Requirements

- Availability: System should be available 24/7.
- Reliability: Accurate tracking and monitoring.
- Scalability: Support multiple buses and users.
- Security: Protect payment and location data.
- Performance: Location updates within a few seconds.
- Usability: Easy-to-use interface for students and administrators.

6. System Components

Hardware:

- GPS-enabled smartphone or IoT tracking device.
- Fuel sensor.
- Seat occupancy sensors (optional).

- Vehicle monitoring sensors.

Software:

- Mobile Application.
- Admin Dashboard.
- Backend Server.
- Database Management System.
- Map Integration API.

7. Use Cases

Student:

- View bus location.
- Check seat availability.
- Make QR payments.

Driver:

- Share GPS location.
- Follow assigned route.

Administrator:

- Monitor buses.
- View speed and fuel reports.
- Manage routes and stops.
- Receive alerts.

8. Constraints

- Internet connectivity required.
- GPS accuracy depends on signal quality.
- Hardware installation costs for sensors and IoT devices.
- Compliance with transportation and data privacy policies.

9. Expected Benefits

- Reduced waiting time for students.
- Improved transportation management.
- Enhanced safety monitoring.
- Better fuel efficiency analysis.
- Increased transparency and operational control.

10. Conclusion

UniTrack BD aims to modernize university transportation by combining real-time tracking, smart payment solutions, seat monitoring, and vehicle management tools into a single integrated platform.